

Easy (10 Questions)

1.

What does NLB stand for?

- A) Network Link Bridge
- B) Network Load Balancing ☒
- C) Network Logical Backup
- D) Network Level Bridge

Answer: B

Explanation: NLB stands for Network Load Balancing, a feature to distribute network traffic.

2.

What is the main purpose of NLB?

- A) Backup data
- B) Load balance network traffic ☒
- C) Encrypt data
- D) Monitor user activity

Answer: B

Explanation: NLB distributes network traffic across multiple servers for better performance.

3.

Which IP address is shared by all nodes in an NLB cluster?

- A) Unique IP address
- B) Virtual IP address (VIP) ☒
- C) Broadcast IP address
- D) Multicast IP address

Answer: B

Explanation: All nodes share a Virtual IP address to appear as one service.

4.

Which NLB mode uses a multicast MAC address?

- A) Unicast
- B) Multicast ☒
- C) Broadcast
- D) Static

Answer: B

Explanation: Multicast mode assigns a multicast MAC address to the NLB cluster.

5.

What Windows tool is commonly used to manage NLB?

- A) Task Manager
- B) NLB Manager ☒
- C) Registry Editor
- D) Device Manager

Answer: B

Explanation: NLB Manager is the GUI tool to configure and monitor NLB clusters.

6.

NLB improves which of the following?

- A) Network bandwidth
- B) Server availability ☒
- C) Disk space
- D) CPU speed

Answer: B

Explanation: NLB improves server availability by distributing load and providing failover.

7.

Which protocol is commonly balanced by NLB?

- A) HTTP ☒

- B) FTP
- C) SMTP
- D) DHCP

Answer: A

Explanation: HTTP traffic is commonly balanced by NLB clusters in web servers.

8.

Which one is NOT a valid NLB affinity setting?

- A) None
- B) Single
- C) Class C
- D) Class A ☒

Answer: D

Explanation: Valid affinity options are None, Single, and Class C; Class A is not valid.

9.

Which of these is a common symptom of NLB failure?

- A) No network traffic at all
- B) One server handling all traffic ☒
- C) Faster response times

D) More free disk space

Answer: B

Explanation: If NLB fails, load balancing may stop and one server may get all traffic.

10.

What is a typical number of servers in an NLB cluster?

- A) One
- B) Two or more
- C) Ten only
- D) Exactly five

Answer: B

Explanation: NLB requires at least two nodes for load balancing.

Medium (20 Questions)

11.

Which network protocol does NLB use for cluster communication?

- A) TCP

- B) UDP ☒
- C) ICMP
- D) FTP

Answer: B

Explanation: NLB uses UDP packets for cluster heartbeat and communication.

12.

In unicast mode, how many MAC addresses do cluster hosts share?

- A) One ☒
- B) Two
- C) Unique MAC addresses
- D) None

Answer: A

Explanation: In unicast mode, all nodes share the same MAC address, causing some limitations.

13.

Why can multicast mode cause network flooding if not properly configured?

- A) Because of broadcast storms
- B) Switches not supporting IGMP snooping ☒
- C) Incorrect IP addresses

D) Firewall blocking traffic

Answer: B

Explanation: Without IGMP snooping, multicast packets flood all switch ports, causing issues.

14.

Which port rules mode allows multiple nodes to respond on the same port?

- A) Single host
- B) Multiple hosts ☒
- C) Disabled
- D) Exclusive

Answer: B

Explanation: Multiple hosts filtering mode allows more than one node to handle traffic on the same port.

15.

What command-line tool can be used to manage NLB?

- A) ipconfig
- B) nlbmgr.exe
- C) nlb.exe ☒
- D) netsh

Answer: C

Explanation: The nlb.exe command-line utility manages NLB clusters.

16.

What happens if the NLB service is stopped on one node?

- A) Node continues processing requests
- B) Node is removed from the cluster ☒
- C) Cluster shuts down
- D) All nodes stop working

Answer: B

Explanation: If NLB service stops, the node leaves the cluster and stops balancing traffic.

17.

Which traffic type is NOT balanced by NLB?

- A) TCP traffic
- B) UDP traffic
- C) ICMP traffic ☒
- D) HTTP traffic

Answer: C

Explanation: NLB generally does not balance ICMP (ping) traffic.

18.

How does NLB handle a node failure?

- A) Reboots the node automatically
- B) Removes the node from the cluster and redistributes traffic ☒
- C) Sends alert only
- D) Stops the entire cluster

Answer: B

Explanation: NLB removes the failed node and redistributes traffic to healthy nodes.

19.

Which of these is NOT a recommended best practice for NLB?

- A) Use static IP addresses
- B) Enable DHCP on cluster nodes ☒
- C) Monitor cluster health
- D) Test failover

Answer: B

Explanation: DHCP is not recommended for cluster nodes to avoid IP conflicts.

20.

How does “Single” affinity affect client connections?

- A) All clients share sessions
- B) Each client IP connects to the same node ☒
- C) All clients connect randomly
- D) Clients connect based on protocol

Answer: B

Explanation: Single affinity ensures a client always connects to the same node by IP.

21.

Which event log category is best to check for NLB errors?

- A) Security
- B) System ☒
- C) Application
- D) Setup

Answer: B

Explanation: System logs contain NLB service events and errors.

22.

Which port must be open on firewalls to allow NLB heartbeat?

- A) TCP 80
- B) UDP 2504 and 2505 ☒
- C) TCP 443
- D) UDP 53

Answer: B

Explanation: NLB heartbeat uses UDP ports 2504 and 2505 for cluster communication.

23.

What is the effect of incorrect port rule configuration?

- A) Improved security
- B) Traffic may be blocked or misrouted ☒
- C) Faster network traffic
- D) No effect

Answer: B

Explanation: Incorrect port rules can block traffic or cause uneven load balancing.

24.

Which NLB feature ensures a user session remains on the same server?

- A) Port rules
- B) Affinity ☒
- C) Unicast mode
- D) Multicast mode

Answer: B

Explanation: Affinity keeps sessions consistent on one node.

25.

What is the purpose of the virtual IP address in NLB?

- A) For management access only
- B) To represent the cluster as a single service endpoint ☒
- C) For multicast communication
- D) To identify physical nodes

Answer: B

Explanation: The VIP is what clients connect to, representing the cluster.

26.

Which of the following is a typical sign of communication failure between NLB nodes?

- A) Nodes are converged

- B) Nodes are in 'converging' state for a long time ☒
- C) Nodes have unique IPs
- D) NLB service is running

Answer: B

Explanation: Long 'converging' indicates cluster nodes cannot communicate properly.

27.

In troubleshooting NLB, what is the purpose of pinging the virtual IP?

- A) To check network connectivity to the cluster ☒
- B) To reset the cluster
- C) To stop traffic
- D) To assign new IP

Answer: A

Explanation: Ping VIP verifies if the cluster is reachable on the network.

28.

Which Windows PowerShell cmdlet checks NLB cluster node status?

- A) Get-ClusterNode
- B) Get-NlbClusterNode ☒
- C) Get-NetAdapter

D) Get-Service

Answer: B

Explanation: Get-NlbClusterNode shows the status of nodes in the NLB cluster.

29.

What is the main disadvantage of using unicast mode?

- A) Nodes cannot communicate directly on the network ☒
- B) Traffic is encrypted
- C) Cluster cannot be managed
- D) It uses multicast MAC addresses

Answer: A

Explanation: In unicast, nodes share MAC address, preventing direct node-to-node communication.

30.

Which mode of NLB is best when the switch supports IGMP snooping?

- A) Unicast
- B) Multicast with IGMP snooping ☒
- C) Broadcast
- D) Static

Answer: B

Explanation: IGMP multicast mode reduces network flooding and is preferred with IGMP support.

Hard (10 Questions)

31.

When NLB is in multicast mode without IGMP snooping, what is the common network problem observed?

- A) Packet loss due to collisions
- B) Broadcast storms flooding the network ☒
- C) Network switch crashes
- D) No connectivity between clients

Answer: B

Explanation: Without IGMP snooping, multicast packets flood all ports causing network congestion.

32.

Which Windows registry key stores NLB cluster configuration?

- A) HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\WLBS ☒
- B) HKEY_CURRENT_USER\Network\NLB
- C) HKEY_LOCAL_MACHINE\Software\Microsoft\NLB

D) HKEY_USERS\NLB

Answer: A

Explanation: The WLBS key holds NLB service configuration.

33.

What is the effect of enabling 'Drainstop' on an NLB node?

- A) Node stops immediately
- B) Node stops receiving new connections but finishes current sessions ☒
- C) Node restarts cluster
- D) Node rejects all traffic

Answer: B

Explanation: Drainstop allows graceful removal of a node by finishing existing sessions first.

34.

How does NLB handle client requests if two nodes have overlapping port rules with 'Multiple hosts' filtering mode?

- A) Only one node responds
- B) Both nodes respond to different clients based on hashing ☒
- C) Requests are blocked
- D) Cluster shuts down

Answer: B

Explanation: NLB distributes traffic based on client IP/port hashing among nodes.

35.

Which scenario may require configuring multicast mode over unicast mode?

- A) Network switch does not support multicast
- B) Nodes must communicate with each other on the network ☒
- C) Cluster has only one node
- D) Using DHCP addresses on nodes

Answer: B

Explanation: Multicast allows direct node-to-node communication, needed for some apps.

36.

How does NLB maintain session persistence with class C affinity?

- A) Binds all client IPs to one node
- B) Routes clients from the same Class C subnet to the same node ☒
- C) Ignores client IP and routes randomly
- D) Routes based on port only

Answer: B

Explanation: Class C affinity groups clients by subnet for session persistence.

37.

Which of the following is a limitation of NLB compared to hardware load balancers?

- A) Supports only TCP traffic
- B) Does not support SSL offloading ☒
- C) Cannot distribute traffic
- D) No fault tolerance

Answer: B

Explanation: NLB does not support SSL termination/offloading like hardware balancers.

38.

In troubleshooting, if the NLB cluster is stuck in converging state, which of the following is the least likely cause?

- A) Network firewall blocking UDP ports ☒
- B) Incorrect subnet mask on nodes
- C) Duplicate IP addresses in cluster
- D) NLB service running on all nodes

Answer: D

Explanation: NLB running on nodes is necessary; it does not cause convergence issues.

39.

What happens if the virtual IP is configured on a node's physical network adapter in addition to its dedicated IP?

- A) Node may not join cluster correctly ☒
- B) Better load balancing
- C) Automatic failover
- D) No effect

Answer: A

Explanation: VIP should be separate from physical IP to avoid IP conflicts and cluster issues.

40.

Which PowerShell cmdlet can be used to add a node to an existing NLB cluster?

- A) Add-NlbClusterNode ☒
- B) New-NlbCluster
- C) Start-NlbCluster
- D) Set-NlbClusterNode

Answer: A

Explanation: Add-NlbClusterNode adds a new node to an existing NLB cluster